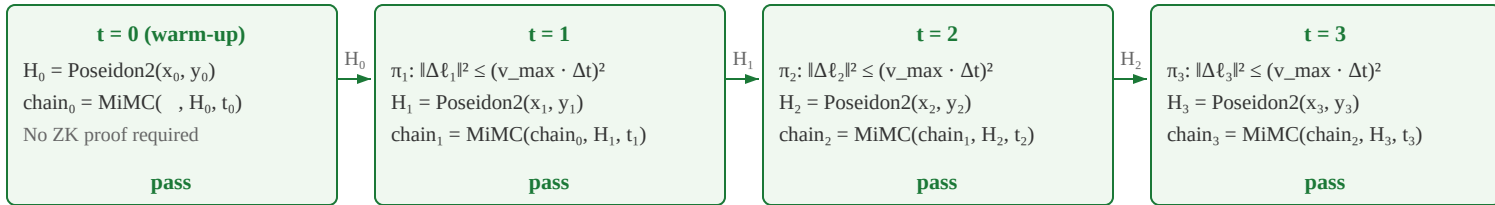
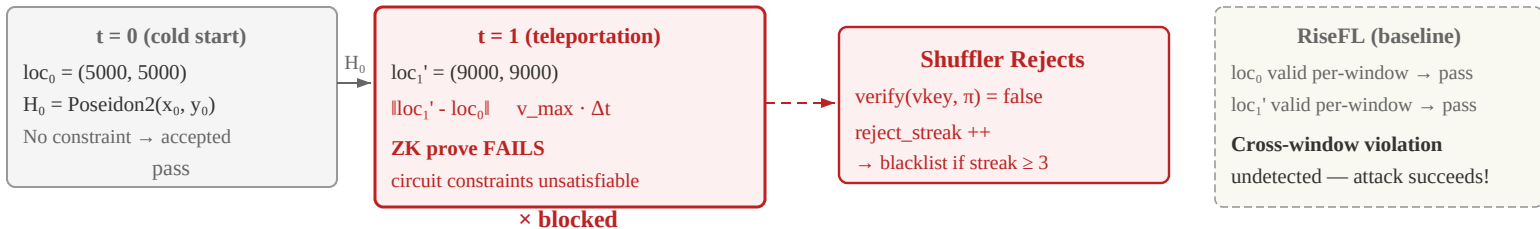


**Fig. 3. TSIP Commitment Chain: Honest User vs. Attacker**

**(a) Honest user — all rounds accepted**



**(b) Attacker — teleportation attack blocked by TSIP**



**Key insight: TSIP binds consecutive positions via Poseidon2 commitments + ZK distance proof.**

Attackers cannot forge a valid proof  $\pi$  when  $\|\Delta \text{loc}\| > v\_max \cdot \Delta t$  (Groth16 soundness:  $\Pr[\text{forge}] \leq 2^{-128}$ ).